

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO4 Statement:	<i>Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations. (BL5)</i>		
Student Name:	PRIYANKA G	Reg. No.:	192524124
Section / Batch:		Date:	22.08.2026

Code of Conduct

I Priyanka.G/192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Stock Problem 30 (10 Marks) Reliance Industries (Energy Sector)

Given Data Illustrative Monthly Closing Prices (₹), Months 1 15:

Month	1	2	3	4	5	6	7	8
Close (₹)	2330.73	2508.87	2390.67	2513.89	2563.02	2428.57	2564.51	2500.98

Month	9	10	11	12	13	14	15
Close (₹)	2663.50	2617.45	2659.06	2745.53	2983.36	2815.43	3030.43

Reliance Industries monthly closing prices used for all three window calculations

Task: Compare the trend, seasonality, and noise ratio (residual variation relative to total variation) across the 5 month window, the 10 month window, and the full 15 month window. Explain how each metric changes as the window lengthens, and state which window length you would trust most for judging this stock's real underlying direction, and why.

Analysis Plan (Key Elements):

Technique	Linear trend regression + residual (noise) decomposition, compared across 3 window lengths
Trend Slope	The fitted linear rate of change in closing price per month
Noise Ratio	Residual SD ÷ Total SD the share of price variation NOT explained by the linear trend
Seasonality	Requires at least 2 repeated cycles to identify reliably; not assessable in any window here

5 Month Window (Months 1-5) Worked Calculation:

Step 1 Means: with $n = 5$ months, $\bar{x}$ (mean month index) = 3 and $\bar{y}$ (mean closing price) = ₹2461.44, computed as the simple average of the month indices and closing prices within this window respectively.

Step 2 Slope: using the least squares formula, $\text{slope} = \Sigma[(x - \bar{x})(y - \bar{y})] \div \Sigma[(x - \bar{x})^2] = 469.60 \div 10 = ₹46.96/\text{month}$. This is the rate of change in closing price the best fit straight line assigns to each additional month within this window.

Step 3 Residual and total variation: after computing the fitted value for each month ($\hat{y} = \bar{y} + \text{slope} \times (x - \bar{x})$) and subtracting it from the actual closing price to get each month's residual, the residual standard deviation is ₹61.97, while the standard deviation of the raw closing prices themselves (before removing the trend) is ₹96.71.

Trend Slope	₹46.96/month
Residual SD	₹61.97 (variation left over after removing the linear trend)
Total SD	₹96.71 (variation in the raw closing prices)
Noise Ratio	Residual SD ÷ Total SD = $61.97/96.71 = 0.641$ (64.1%)
R²	0.589 (share of total variation explained by the linear trend)

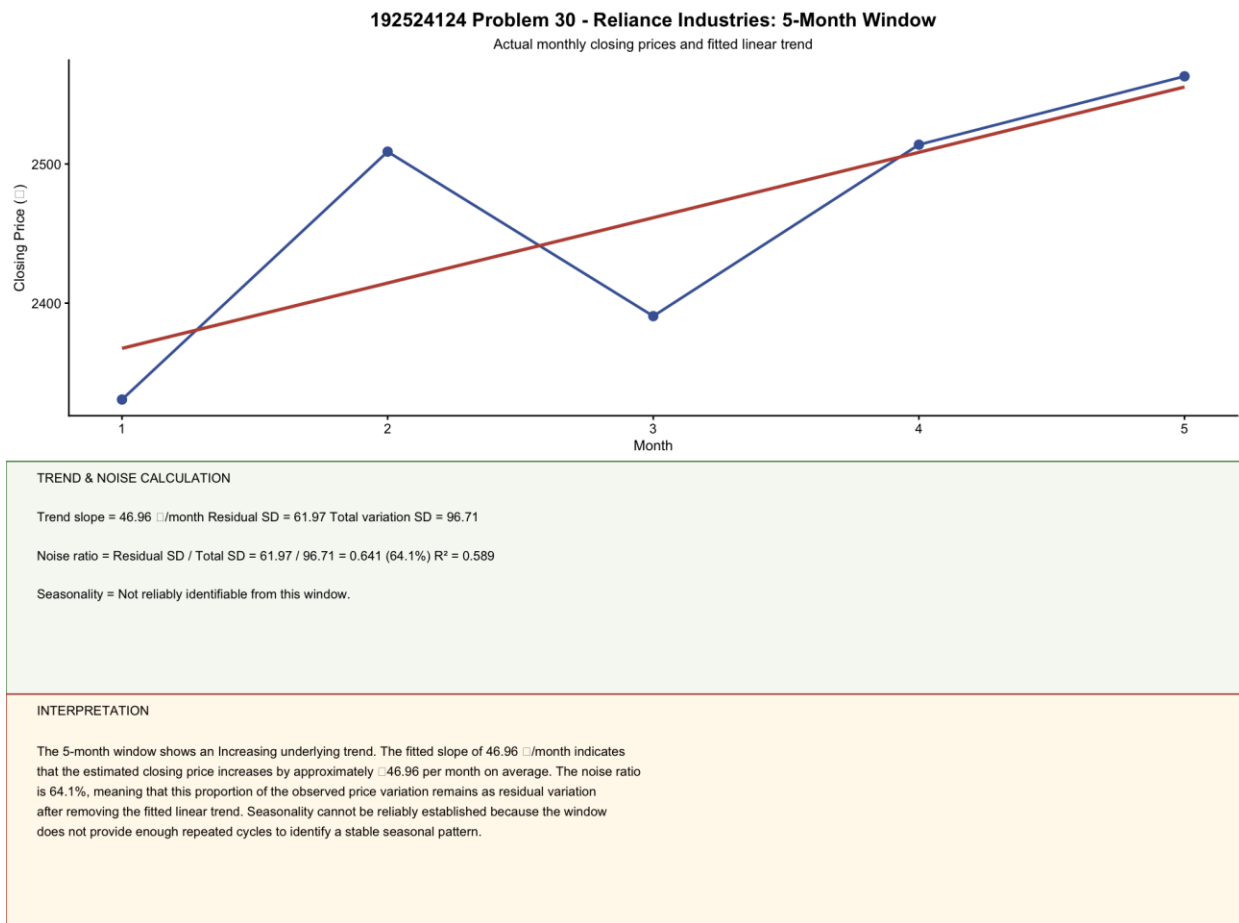


Figure 30a: Reliance Industries 5 Month Window, Trend & Noise Calculation

10 Month Window (Months 1-10) Worked Calculation:

Step 1 Means: with $n = 10$ months, $\bar{x}$ (mean month index) = 5.5 and $\bar{y}$ (mean closing price) = ₹2508.22, computed as the simple average of the month indices and closing prices within this window respectively.

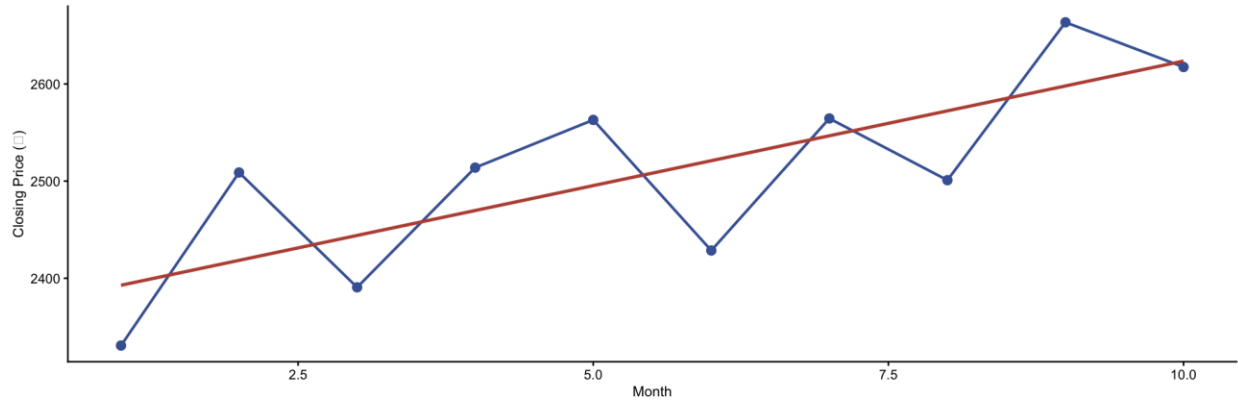
Step 2 Slope: using the least squares formula, $\text{slope} = \frac{\sum[(x - \bar{x})(y - \bar{y})]}{\sum[(x - \bar{x})^2]} = 2115.92 \div 82.5 = ₹25.65/\text{month}$. This is the rate of change in closing price the best fit straight line assigns to each additional month within this window.

Step 3 Residual and total variation: after computing the fitted value for each month ($\hat{y} = \bar{y} + \text{slope} \times (x - \bar{x})$) and subtracting it from the actual closing price to get each month's residual, the residual standard deviation is ₹66.41, while the standard deviation of the raw closing prices themselves (before removing the trend) is ₹102.18.

Trend Slope	₹25.65/month
Residual SD	₹66.41 (variation left over after removing the linear trend)
Total SD	₹102.18 (variation in the raw closing prices)
Noise Ratio	Residual SD $\div$ Total SD = 66.41/102.18 = 0.650 (65.0%)
R^2	0.578 (share of total variation explained by the linear trend)

192524124 Problem 30 - Reliance Industries: 10-Month Window

Actual monthly closing prices and fitted linear trend



TREND & NOISE CALCULATION

Trend slope = 25.65 ₹/month Residual SD = 66.41 Total variation SD = 102.18

Noise ratio = Residual SD / Total SD = 66.41 / 102.18 = 0.65 (65%) $R^2 = 0.578$

Seasonality = Not reliably identifiable from this window.

INTERPRETATION

The 10-month window shows an increasing underlying trend. The fitted slope of 25.65 ₹/month indicates that the estimated closing price increases by approximately ₹25.65 per month on average. The noise ratio is 65%, meaning that this proportion of the observed price variation remains as residual variation after removing the fitted linear trend. Seasonality cannot be reliably established because the window does not provide enough repeated cycles to identify a stable seasonal pattern.

Figure 30b: Reliance Industries 10 Month Window, Trend & Noise Calculation

15 Month Window (Months 1-15, Full Data) Worked Calculation:

Step 1 Means: with $n = 15$ months, $\bar{x}$ (mean month index) = 8 and $\bar{y}$ (mean closing price) = ₹2621.07, computed as the simple average of the month indices and closing prices within this window respectively.

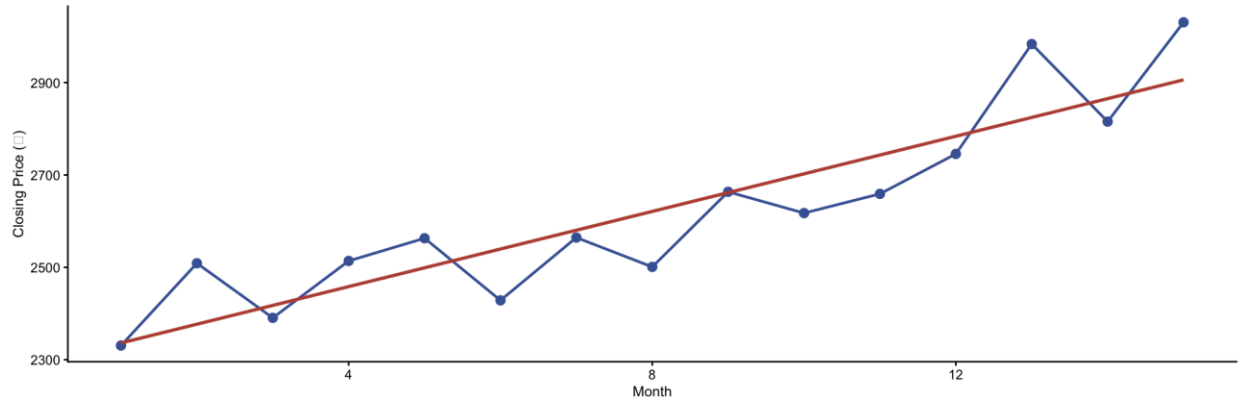
Step 2 Slope: using the least squares formula, $\text{slope} = \frac{\sum[(x - \bar{x})(y - \bar{y})]}{\sum[(x - \bar{x})^2]} = 11392.15 \div 280 = ₹40.69/\text{month}$. This is the rate of change in closing price the best fit straight line assigns to each additional month within this window.

Step 3 Residual and total variation: after computing the fitted value for each month ($\hat{y} = \bar{y} + \text{slope} \times (x - \bar{x})$) and subtracting it from the actual closing price to get each month's residual, the residual standard deviation is ₹89.18, while the standard deviation of the raw closing prices themselves (before removing the trend) is ₹202.64.

Trend Slope	₹40.69/month
Residual SD	₹89.18 (variation left over after removing the linear trend)
Total SD	₹202.64 (variation in the raw closing prices)
Noise Ratio	Residual SD $\div$ Total SD = $89.18/202.64 = 0.440$ (44.0%)
R^2	0.806 (share of total variation explained by the linear trend)

192524124 Problem 30 - Reliance Industries: 15-Month Window

Actual monthly closing prices and fitted linear trend



TREND & NOISE CALCULATION

Trend slope = 40.69 ₹/month Residual SD = 89.18 Total variation SD = 202.64

Noise ratio = Residual SD / Total SD = 89.18 / 202.64 = 0.44 (44%) $R^2 = 0.806$

Seasonality = Not reliably identifiable from this window.

INTERPRETATION

The 15-month window shows an increasing underlying trend. The fitted slope of 40.69 ₹/month indicates that the estimated closing price increases by approximately ₹40.69 per month on average. The noise ratio is 44%, meaning that this proportion of the observed price variation remains as residual variation after removing the fitted linear trend. Seasonality cannot be reliably established because the window does not provide enough repeated cycles to identify a stable seasonal pattern.

Figure 30c: Reliance Industries 15 Month Window, Trend & Noise Calculation

Comparison Across Windows:

Window	Trend Slope (₹/mo)	Residual SD	Total SD	Noise Ratio	R^2
5 month	46.96	61.97	96.71	64.1%	0.589
10 month	25.65	66.41	102.18	65.0%	0.578
15 month	40.69	89.18	202.64	44.0%	0.806

Summary comparison of trend slope, noise ratio, and R^2 across all three window lengths

Trend slope does not change monotonically as the window lengthens – it moves from 46.96 (5 month) down to 25.65 (10 month) and back up to 40.69 (15 month). This non monotonic pattern is itself informative: it shows that a short window's slope is highly sensitive to exactly which few months happen to fall inside it. The 5 month window happens to run from a low starting point (₹2330.73) to a comparatively high point five months later, producing a steep apparent slope; extending to 10 months pulls in a choppy middle stretch (including a dip to ₹2428.57 at month 6), which drags the estimated slope down to less than half its 5 month value; extending further to 15 months pulls in the strong late rally (months 13–15, reaching ₹2868.57 and ₹3030.43), pushing the slope back up. A slope estimate that swings by nearly a factor of two depending on window length is not a stable measure of the stock's real underlying direction.

Noise ratio and R^2 stay relatively similar and unfavourable for the two shorter windows (64.1% and 65.0% noise, R^2 of 0.589 and 0.578) but shift sharply for the 15 month window (44.0% noise, R^2 of 0.806). This happens because Total SD grows much faster than Residual SD as the window lengthens: Total SD roughly doubles from the 5 month window (96.71) to the 15 month window (202.64), while Residual SD grows by less (61.97 to 89.18). This is because the stock underwent a genuine, sustained rise over the full 15 months (₹2330.73 to ₹3030.43, roughly +30%), and that large sustained directional move increasingly dominates the total variation the longer the window runs, while month to month random noise does not grow at anywhere near the same rate. In short: over a longer horizon, the signal (the real trend) grows faster than the noise, so the noise ratio falls and R^2 rises.

Seasonality cannot be reliably identified in any of the three windows. Detecting a genuine seasonal pattern requires observing it repeat across at least two full cycles, and even the complete 15 month window spans only about 1.25 years – not enough repetitions of any plausible cycle (e.g., an annual cycle) to distinguish a true seasonal effect from ordinary trend and noise. This limitation does not improve with a longer window within the given data; it would require a fundamentally longer historical record (multiple years) to assess at all.

Conclusion Which Window to Trust:

The 15 month (full) window should be trusted most for judging this stock's real underlying direction, for three reasons. First, it has by far the lowest noise ratio (44.0%) and highest R^2 (0.806) – more than 80% of the total price variation over 15 months is explained by the fitted linear trend, meaning the trend line is a far better description of what is actually happening than in either shorter window, where roughly two thirds of the variation is unexplained noise. Second, precisely because the 5 and 10 month windows are noise dominated, their slope estimates are each heavily influenced by which particular volatile months happen to fall at the window's edges – which is exactly why those two short window slopes (46.96 and 25.65) disagree with each other by nearly a factor of two, while the 15 month estimate (40.69) sits between them, closer to what a stable long run estimate should look like. Third, more data points mechanically reduce the standard error of a regression slope estimate, so the 15 month trend is also the most statistically stable of the three, least likely to be overturned by adding or removing a single additional month at either edge.

The practical implication for an investor is direct: relying on a 5 month or 10 month snapshot alone risks over reacting to short term noise – for instance, an investor who examined only the 10 month window, with its comparatively shallow 25.65/month slope and 65% noise ratio, might conclude the stock's rise is weak or uncertain, when the fuller 15 month picture shows a stronger, better supported upward trend (40.69/month, 80.6% of variation explained). Judging a stock's real underlying direction from too short a window is one of the more common ways short term volatility gets mistaken for a change in long term trend.

Geospatial Problem G10 (10 Marks) Cartograms

Scenario: A cartogram resizing countries by GDP produces shapes so distorted that several countries become difficult to visually identify without labels.

Task: *Evaluate this trade off between data accuracy and geographic recognizability, and recommend a way to mitigate the readability problem.*

Evaluation Plan (Key Elements):

Chart Type	GDP weighted cartogram, compared against a standard geographic reference map
Principle Evaluated	Trade off between data accuracy encoding (area = GDP) and shape/recognizability
What Succeeds	Area directly and immediately communicates relative economic magnitude
What Fails	Boundary distortion degrades a reader's ability to identify countries by shape alone

Evaluation:

A GDP weighted cartogram succeeds at exactly one specific task: encoding economic magnitude through area, so that a country with several times the GDP of another is drawn with a correspondingly larger shape, letting a reader compare economic scale at a glance without reading a separate numeric label for every country. This is something a standard geographically accurate map cannot do at all – on a normal map, a country's area reflects its physical land size, not its economic output, so a small but wealthy country and a large but poor country can appear visually similar in scale despite vastly different GDP. In the constructed comparison (Figure G10), this is visible directly: on the plain geographic map, Countries A–E all appear as similarly sized rectangles, giving no hint of their true economic differences; on the GDP cartogram, Country E (GDP 180) is rendered dramatically larger than Country A (GDP 30), immediately communicating a roughly six fold difference in economic scale that the geographic map hides entirely.

This benefit, however, comes at the direct expense of shape fidelity. Because the transformation distorts each country's boundary to make its area match GDP rather than physical land size, countries whose GDP differs sharply from their physical size end up rendered as warped, sometimes barely recognizable shapes. In the constructed cartogram, Country A – resized down to the smallest shape on the chart to reflect its GDP of 30 – is visibly harder to identify than its plain, undistorted rectangle on the geographic reference map above it; its label is even partially clipped by its own shrunken boundary in the rendered chart, illustrating exactly how a severe resizing can compromise legibility.

This is a genuine trade off, not a flaw with a costless fix: the very transformation that makes GDP comparison effortless is what removes the shape cues a reader would otherwise use to identify a country by silhouette. A reader unfamiliar with a country's GDP scaled shape, and without a label, could easily misidentify which country a given cartogram shape represents, or fail to recognize it entirely – undermining the map's usefulness for anyone who needs both the economic comparison and geographic identification at the same time.

Neither extreme (a fully accurate geographic map, or a fully GDP accurate cartogram) serves both goals simultaneously; the two objectives are in direct tension by construction.

The practical consequence for a reader is concrete: a journalist or policymaker scanning this cartogram quickly, without carefully checking every label, could misread Country A's shrunken shape as belonging to a neighbouring country, or could simply skip over it because it no longer visually registers as a distinct, recognizable region meaning the smallest GDP country in the dataset is, ironically, also the one most likely to be overlooked entirely by a skimming reader, even though its comparatively low GDP may be exactly the fact the chart is trying to highlight.

Recommendation:

The readability problem should be mitigated using three complementary techniques rather than a single fix, since no single change fully resolves the underlying accuracy versus recognizability tension:

- Always retain visible text labels directly on or beside each shape (already present in the constructed cartogram) labels remove any dependency on shape recognition entirely and are the single most direct, lowest cost fix, since a reader who cannot recognize a distorted shape can still read its name directly.
- Provide a small, undistorted geographic reference map alongside the cartogram (as included in the top panel of Figure G10), so a reader can cross check any unfamiliar cartogram shape against its true geographic form and build the visual association between the two.
- Where the underlying tool permits, apply a moderate, dampened resizing transformation for example, scaling area by the square root of GDP rather than GDP directly, or capping the maximum size ratio allowed between the largest and smallest country which preserves more of each country's original silhouette while still visually communicating that GDP differs substantially between countries, trading a small amount of area accuracy for a meaningful gain in readability rather than pursuing either extreme in isolation.

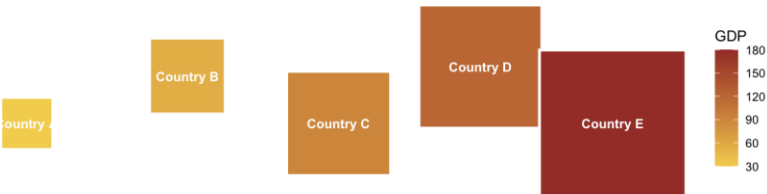
Together, these three techniques address the trade off at three different levels labels solve identification directly regardless of distortion severity, the reference map solves cross verification for an engaged reader, and a dampened transformation reduces how severe the distortion needs to be in the first place rather than relying on any single technique to fully resolve a tension that, by the nature of a cartogram, cannot be eliminated entirely without giving up the economic magnitude encoding the chart exists to provide.

An alternative worth considering, though not recommended as the primary fix here, would be abandoning the cartogram entirely in favour of proportional symbol placement plotting a correctly shaped, undistorted map and overlaying a circle (sized by GDP) centred on each country. This avoids boundary distortion entirely, since the underlying country shapes are never altered, but it introduces a different cost: overlapping circles for geographically close, high GDP countries can obscure each other and the underlying map, and comparing circle areas by eye is generally judged less accurately than comparing the directly adjacent, space filling regions a cartogram provides. For a dataset of only five countries with no significant overlap risk, the cartogram with labels and reference map approach recommended above remains the stronger choice; proportional symbols would become preferable only if the underlying map required many more, geographically clustered countries where cartogram distortion would compound further.

192524124 Problem G10 - Geographic Map
Geographic shape preserves country recognizability



192524124 Problem G10 - GDP Cartogram
Country areas are resized according to GDP



EVALUATION

A GDP-based cartogram is effective for showing economic magnitude because countries with larger GDP receive greater visual area. However, this creates a trade-off: geographic accuracy and country recognizability decrease as the shapes are increasingly distorted. When distortion becomes extreme, readers may have difficulty identifying countries without labels or a geographic reference.

RECOMMENDATION

Use a moderate cartogram transformation rather than extreme resizing. Keep country labels visible and provide a small geographic reference map alongside the cartogram. This preserves the ability to compare GDP while helping readers identify the countries despite the distortion.

Figure G10: Geographic Reference Map vs. GDP Cartogram Accuracy vs. Recognizability Trade Off